

60016 OPERATIONS RESEARCH

Game Theory (CW)

Lecture 09

Game Theory

Optimization (e.g. LP) → problems with a **single decision-maker**:

- ▶ LP: diet optimisation, newsvendor problem, ...
- ▶ ILP: supply chain design, airline crew scheduling, ...

Game theory → problems with **multiple decision-makers**

Game theoretical inference of human behavior in social networks

Nicolò Pagan  & Florian Dörfler

Nature Communications 10, Article



Pacing Equilibrium in First-Price Auction Markets

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Real-Time Bidding



Incentivizing efficient use of shared infrastructure: Optimal tolls in congestion games

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Games

Key ingredients:

- ▶ Every player takes own decision
- ▶ Payoff each player receives depends on choice of all players

In “math language”:

- ▶ Each player $i = 1, \dots, n$ has a set of actions $x_i \in \mathcal{X}_i$
- ▶ Player i receives a payoff $J_i(x_1, \dots, x_i, \dots, x_n)$

Very special class: two-Person Zero-Sum Games w finite actions

- ▶ Two players: **row player** (RP) and **column player** (CP)
- ▶ RP chooses one out of m strategies (**row strategies**)
- ▶ CP chooses one out of n strategies (**column strategies**)
- ▶ **Zero-Sum**: RP wins whatever CP loses and viceversa

Equilibrium notions

- ▶ Dominant equilibrium
- ▶ Pure Nash equilibrium
- ▶ Mixed Nash equilibrium

Payoff Matrix

- ▶ **Payoff matrix**: descriptor of a two-player zero-sum game
- ▶ If RP plays strategy i and CP plays strategy j , then CP pays a_{ij} to RP

		<i>CP</i>			
		Strategy 1	Strategy 2	...	Strategy n
<i>RP</i>	Strategy 1	a_{11}	a_{12}	...	a_{1n}
	Strategy 2	a_{21}	a_{22}	...	a_{2n}
	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
	Strategy m	a_{m1}	a_{m2}	...	a_{mn}

Example: Odds-and-Evens

- ▶ Both players simultaneously show “1” or “2” fingers
- ▶ If the sum of both numbers is even: CP gives £1 to RP
- ▶ If the sum of both numbers is odd: RP gives £1 to CP

		<i>CP</i>	
		1 Finger	2 Fingers
<i>RP</i>	1 Finger	1	-1
	2 Fingers	-1	1

Two-Person Zero-Sum Games

Assumptions of Two-Person Zero-Sum Games:

1. Each player knows the **game setting** (available strategies to RP and CP, values of payoff matrix)
2. Both players **simultaneously choose** their strategy, that is, **without knowing** what their opponent chooses
3. Each player chooses a strategy that enables him/her to do best, reasoning as if the opponent could anticipate his/her strategy
4. Both players are **rational**:
 - ▶ They try to maximise their utility
 - ▶ They show no compassion for their opponent

Running Example

Elections game:

1. Two players: RP (row) and CP (column)
2. Both players have three strategies:
 - ▶ L: campaign the last two days in London
 - ▶ B: campaign the last two days in Birmingham
 - ▶ S: split the last two days, campaign one day in London and one day in Birmingham
3. **Payoffs:** how many voters does RP acquire from CP ?

Dominant equilibrium

Running Example

Consider the following setting of the Elections Game:

		<i>CP</i>		
		L	B	S
<i>RP</i>	L	1	2	4
	B	1	0	5
	S	0	1	-1

We want to find the strategies that will be played by RP and CP.

Running Example

		<i>CP</i>		
		L	B	S
<i>RP</i>	L	1	2	4
	B	1	0	5
	S	0	1	-1

Observation:

Strategy L (“London only”) is always better for RP than Strategy S (“London and Birmingham”).

Conclusion:

RP will never play strategy S. Both players will realise this and we can ignore it.

Running Example

		<i>CP</i>		
		L	B	S
<i>RP</i>	L	1	2	4
	B	1	0	5

Observation:

Both strategy L (“London only”) and strategy B (“Birmingham only”) are always better (less to pay) for CP than strategy S (“Split between London and Birmingham”).

Conclusion:

CP will never play strategy S. Both players will realise this and we can ignore it.

Running Example

		<i>CP</i>	
		L	B
<i>RP</i>	L	1	2
	B	1	0

Observation:

Strategy L (“London only”) is no worse for RP than strategy B (“Birmingham only”) and can be better (if CP plays B).

Conclusion:

RP will never play strategy B. Both players will realise this and we can ignore it.

Running Example

		<i>CP</i>	
		L	B
<i>RP</i>	L	1	2

Observation:

Strategy L (“London only”) is always better for CP than strategy B (“Birmingham only”).

Conclusion:

CP will never play strategy B. Both players will realise this and we can ignore it.

Running Example

		<i>CP</i>	
		L	
<i>RP</i>	L	<table><tr><td>1</td></tr></table>	1
1			

Dominant Strategy Equilibrium:

Both RP and CP will campaign in London.

Dominance

Dominated row strategy: Row strategy i is dominated by row strategy i' if $a_{i'j} \geq a_{ij}$ for all column strategies $j = 1, \dots, n$ and $a_{i'j} > a_{ij}$ for some j .

Dominated column strategy: Column strategy j is dominated by column strategy j' if $a_{ij'} \leq a_{ij}$ for all row strategies $i = 1, \dots, m$ and $a_{ij'} < a_{ij}$ for some i .

- ▶ A rational player will never play a dominated strategy
- ▶ A rational opponent knows this

Dominant Strategy Equilibria

Dominant Strategy Equilibrium: If a repeated removal of dominated strategies leads to a game where each player has just one strategy left, then this strategy pair is a dominant strategy equilibrium.

Properties:

- ▶ If a dominant strategy equilibrium exists, then it is unique.
- ▶ If a dominant strategy equilibrium exists, then any rational players will play the associated equilibrium strategies.

Pure Nash equilibrium

Running Example

Consider a different payoff matrix for the Elections Game:

		<i>CP</i>		
		L	B	S
<i>RP</i>	L	-3	-2	6
	B	2	0	2
	S	5	-2	-4

- ▶ This game has no dominated strategies
- ▶ Hence, there is no dominant strategy equilibrium

Security strategy over rows

Assumption: “Each player chooses a strategy that enables him/her to do best in face of worst-case opponent”

- ▶ α_i : payoff of row strategy i , when facing worst-case opponent

$$\alpha_i = \min_{j=1,\dots,n} a_{ij}$$

- ▶ Thus, the RP will pick the strategy i that maximizes the worst-case payoff

$$\max_{i=1,\dots,n} \min_{j=1,\dots,n} a_{ij}$$

Example: security strategy (rows)

		CP			α_i
		L	B	S	
RP	L	-3	-2	6	-3
	B	2	0	2	0
	S	5	-2	-4	-4

- Strategy B is the best for the RP

Security strategy over columns

- ▶ We repeat the reasoning for the CP.
- ▶ β_j : cost of column strategy j , when facing worst-case opponent

$$\beta_j = \max_{i=1,\dots,n} a_{ij}$$

- ▶ Thus, the CP will pick the strategy j that minimizes the worst-case cost

$$\min_{j=1,\dots,n} \max_{i=1,\dots,n} a_{ij}$$

Example: Security strategy (columns)

		<i>CP</i>		
		L	B	S
<i>RP</i>	L	-3	-2	6
	B	2	0	2
	S	5	-2	-4
β_j		5	0	6

- Strategy B is also best for the CP

Nash Equilibrium in Pure Strategies

Definition: a **Nash Equilibrium** is a strategy pair (i^*, j^*) such that no player has an incentive to **unilaterally** deviate from his/her chosen strategy if told the strategy of the other player.

Note: a Nash equilibrium may not always exist in pure strategies.

Properties:

- ▶ If (i^*, j^*) is a Nash equilibrium, then $\alpha_{i^*} = \beta_{j^*}$
- ▶ The payoff of the Nash equilibrium's strategy pair $\alpha_{i^*} = \beta_{j^*}$ is called the **value of the game**.

Mixed Nash equilibrium

Example: Odds-and-Evens (towards mixed strategies)

		<i>CP</i>	
		1	2
<i>RP</i>	1	-1	1
	2	1	-1

- ▶ Players **randomly** pick strategy with equal probabilities
- ▶ Each strategy pair is played with probability 0.25
- ▶ **Expected value of the game** is 0 for both players
 - ▶ No reason to unilaterally change probabilities
 - ▶ Example of **Nash equilibrium in mixed strategies**

Mixed Strategies

- ▶ In a **mixed strategy** $(p_1, \dots, p_m; q_1, \dots, q_n)$:
 - ▶ RP plays strategy i with probability p_i , $\sum_{i=1}^m p_i = 1$.
 - ▶ CP plays strategy j with probability q_j , $\sum_{j=1}^n q_j = 1$.
- ▶ If $p_k = 1$ or $q_k = 1$, then k is a pure strategy
- ▶ The payoff of the mixed strategy (p, q) will be

$$V(p, q) = \sum_{i=1}^m \sum_{j=1}^n p_i q_j a_{ij}$$

- ▶ RP seeks probabilities that maximise payoff $(p_1^*, \dots, p_m^*)$
- ▶ CP seeks probabilities that minimise loss $(q_1^*, \dots, q_n^*)$

Definition

A mixed Nash equilibrium is a pair of mixed strategies (p^*, q^*) such that $V(p, q^*) \leq V(p^*, q^*) \leq V(p^*, q)$ for all other mixed strategies (p, q) [i.e., no agent has any incentive in unilaterally deviating].

Example: Election Game (revised in mixed strategies)

		<i>CP</i>			
		L	B	S	
<i>RP</i>	L	0	-1	2	p_L
	B	5	4	-3	p_B
	S	2	3	-4	p_S
		q_L	q_B	q_S	

- ▶ The strategy pair (S,B) is played with probability $p_S q_B$
- ▶ (S,B) contributes $p_S q_B \cdot 3$ to the mixed strategy payoff
- ▶ How can player find their optimal probabilities?

Column Player's Perspective

- ▶ Remember the **Assumption**: “Each player chooses a strategy that enables him/her to do best, reasoning in face of the worst-case opponent”
- ▶ CP expects RP to respond with optimal p_i 's for any choice of q_j 's. How should CP choose the q_j 's?

$$V_{CP} = \min_{q_1, \dots, q_n} \max_{p_1, \dots, p_m} \sum_{i=1}^m \sum_{j=1}^n p_i q_j a_{ij}$$

subject to

$$\begin{aligned} \sum_{j=1}^n q_j &= 1, & \sum_{i=1}^m p_i &= 1, \\ q_j &\geq 0, & p_i &\geq 0 \end{aligned}$$

Column Player's Perspective (inner program)

Goal: reduce optimisation problem to linear program

- ▶ Let us rewrite and focus on the inner problem

$$V_{CP} = \min_{q_1, \dots, q_n} V_{CP}^n(q_1, \dots, q_n)$$

subject to

$$\sum_{j=1}^n q_j = 1, \quad ,$$
$$q_j \geq 0,$$

Column Player's Perspective (inner program is trivial!)

- ▶ For any choice of q_j 's, let $\alpha_i = \sum_{j=1}^n q_j a_{ij}$ be row payoffs
- ▶ Then the inner maximisation problem is:

$$V_{CP}^{in}(q_1, \dots, q_n) = \max_{p_1, \dots, p_m} \sum_{i=1}^m p_i \alpha_i$$

subject to

$$\sum_{i=1}^m p_i = 1,$$

$$p_i \geq 0$$

- ▶ The solution is $p_i = 1$ for the largest α_i , $p_k = 0$ for $k \neq i$.
- ▶ Example: maximise $3p_1 + 2p_2 + 5p_3 \Rightarrow p_3 = 1$.

Example 1: inner program is trivial

- ▶ CP evaluates a pure strategy $q_S = 1$

		CP			
		L	B	S	
RP	L	0	-1	2	p_L
	B	5	4	-3	p_B
	S	2	3	-4	p_S
		0.0	0.0	1.0	

- ▶ if RP plays L, $\alpha_L = 0.0 \times 0 + 0.0 \times -1 + 1.0 \times 2 = 2$
- ▶ if RP plays B, $\alpha_B = 0.0 \times 5 + 0.0 \times 4 + 1.0 \times -3 = -3$
- ▶ if RP plays S, $\alpha_S = 0.0 \times 2 + 0.0 \times 3 + 1.0 \times -4 = -4$
- ▶ RP optimal response to CP is $p_L = 1 \Rightarrow V_{CP}^{in} = 2$

Example 2: inner program is trivial

- ▶ CP changes guess and evaluates a mixed strategy

		CP			
		L	B	S	
RP	L	0	-1	2	p_L
	B	5	4	-3	p_B
	S	2	3	-4	p_S
		0.7	0.2	0.1	

- ▶ if RP plays L, $\alpha_L = 0.7 \times 0 + 0.2 \times -1 + 0.1 \times 2 = 0$
- ▶ if RP plays B, $\alpha_B = 0.7 \times 5 + 0.2 \times 4 + 0.1 \times -3 = 4$
- ▶ if RP plays S, $\alpha_S = 0.7 \times 2 + 0.2 \times 3 + 0.1 \times -4 = 1.6$
- ▶ RP optimal response to CP is $p_B = 1 \Rightarrow V_{CP}^{in} = 4$

Column Player (substitute inner in outer program)

- ▶ The inner maximisation optimal value is thus simply

$$V_{CP}^{in}(q_1, \dots, q_n) = \max \{\alpha_1, \dots, \alpha_m\}$$

- ▶ Expanding the definitions of the α_i 's, we conclude that CP is in fact solving a min-max problem

$$V_{CP} = \min_{q_1, \dots, q_n} \max \left\{ \sum_{j=1}^n q_j a_{1j}, \dots, \sum_{j=1}^n q_j a_{mj} \right\}$$

subject to

$$\sum_{j=1}^n q_j = 1,$$

$$q_j \geq 0$$

Column Player (final LP)

- ▶ The min-max problem is equivalent to a linear program

$$V_{CP} = \min_{\tau, q_1, \dots, q_n} \tau$$

subject to

$$\tau \geq \sum_{j=1}^n q_j a_{ij}, \quad \forall i = 1, \dots, m$$

$$\sum_{j=1}^n q_j = 1,$$

$$q_j \geq 0,$$

- ▶ Election Game: $q_L^* = 0, q_B^* = \frac{1}{2}, q_S^* = \frac{1}{2} \Rightarrow V_{CP}^* = \frac{1}{2}$
- ▶ Note: the optimal q_j^* 's are independent of the p_i^* 's

Row Player's Perspective

- ▶ A similar reasoning applies to the row player, who instead optimises

$$V_{RP} = \max_{p_1, \dots, p_m} \min_{q_1, \dots, q_n} \sum_{i=1}^m \sum_{j=1}^n p_i q_j a_{ij}$$

subject to

$$\begin{aligned} \sum_{i=1}^m p_i &= 1, & \sum_{j=1}^n q_j &= 1, \\ p_i &\geq 0, & q_j &\geq 0 \end{aligned}$$

Row Player's Perspective

- ▶ The max-min problem can be shown equivalent to a linear program

$$V_{RP} = \max_{\tau, p_1, \dots, p_m} \tau$$

subject to

$$\tau \leq \sum_{i=1}^m p_i a_{ij}, \quad \forall j = 1, \dots, n$$

$$\sum_{i=1}^m p_i = 1,$$

$$p_i \geq 0,$$

- ▶ Election Game: $p_L^* = \frac{7}{10}, p_B^* = \frac{3}{10}, p_S^* = 0 \Rightarrow V_{RP}^* = \frac{1}{2}$
- ▶ Observation: p_i^* 's will be independent of the q_j^* 's.

Minimax Theorem

Theorem (Von Neumann, 1928). For every two-person zero-sum game, the RP and CP linear programs have the same optimal value, i.e.,

$$V_{RP} = \max_{p_1, \dots, p_m} \min_{q_1, \dots, q_n} \sum_{i=1}^m \sum_{j=1}^n p_i q_j a_{ij} = \min_{q_1, \dots, q_n} \max_{p_1, \dots, p_m} \sum_{i=1}^m \sum_{j=1}^n p_i q_j a_{ij} = V_{CP}$$

Further a (p^*, q^*) is a mixed Nash equilibrium.

Proof: ideas?

Minimax Theorem

Theorem (Von Neumann, 1928). For every two-person zero-sum game, the RP and CP linear programs have the same optimal value, i.e.,

$$V_{RP} = \max_{p_1, \dots, p_m} \min_{q_1, \dots, q_n} \sum_{i=1}^m \sum_{j=1}^n p_i q_j a_{ij} = \min_{q_1, \dots, q_n} \max_{p_1, \dots, p_m} \sum_{i=1}^m \sum_{j=1}^n p_i q_j a_{ij} = V_{CP}$$

Further a (p^*, q^*) is a mixed Nash equilibrium.

Proof: ideas? Result follows by **strong duality** since the two programs are the **duals** of each other.

Consequences:

- ▶ A **Nash Equilibrium in mixed strategies always exists!!!**
 - ▶ Players **expect identical payoffs**
 - ▶ Neither player has an incentive to change p_i or q_j
- ▶ Statement generalises to M players (Nash, 1949).

Mixed Nash equilibria - historical note

- Fact: mixed NE are guaranteed to exist (Von Neumann 1928). He later wrote:

"As far as I can see, there could be no theory of games ...without that theorem ...I thought there was nothing worth publishing until the Minimax Theorem was proved."

- ▶ In 1949, Nash gave a **one-page** proof (in 27-page thesis) that games with any number of players have a mixed equilibria.

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[illegible]

times and close the space of singular, $\omega = 0$, is a cell, there must
 be a point ω for some λ . These points will be a continuous, $\omega = 0$,
 connected to $\omega = 0$ and $\omega = 0$, on the right side the point $\omega = 0$, $\omega = 0$,
 The second $\omega = 0$ were not on the right side the point $\omega = 0$, $\omega = 0$,
 $\omega = 0$, $\omega = 0$, $\omega = 0$, some sequence $\omega = 0$, $\omega = 0$,
 and some on the other side, where some $\omega = 0$ were not on the right side
 The same is continuous, $\omega = 0$, $\omega = 0$, $\omega = 0$, $\omega = 0$, $\omega = 0$, $\omega = 0$,

$$P(\omega, \omega) = P(\omega, \omega)$$
 when $\omega = 0$,
 writing

$$P(\omega, \omega) = P(\omega, \omega) < -E$$

 For continuity if ω is large enough

$$|P(\omega, \omega) - P(\omega, \omega)| = |P(\omega, \omega) - P(\omega, \omega)| < \frac{1}{2} \omega$$
 which is
 sufficient

$$P(\omega, \omega) = P(\omega, \omega) + \frac{1}{2} \omega < 0$$
 which is
 enough

$$P(\omega, \omega) = P(\omega, \omega) < 0$$
 because $P(\omega, \omega) = P(\omega, \omega)$

$$C(\omega, \omega) = 0$$
 from the last equation we have that
 The is not set in ω since

$$P(\omega, \omega) = P(\omega, \omega)$$
 because $\omega = 0$
 final point

$$P(\omega, \omega) = P(\omega, \omega)$$
 is not set in ω
 when continuous, $\omega = 0$, $\omega = 0$, $\omega = 0$, $\omega = 0$, $\omega = 0$, $\omega = 0$,
 $\omega = 0$ is not set on multiple points.

- ▶ In 1994, Nash was awarded the Nobel Prize for this work